RESOLVING A CATEGORY ERROR: DHL$[3,2]$ YIELDS $H_1 \le 6$, NOT $H_1 = 2$

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ABSTRACT. Let $H_1 = \liminf_{n\to\infty}(p_{n+1}-p_n)$. We record the precise logical relationship between the hypothesis DHL$[3,2]$ and bounds on H_1. We prove that DHL$[3,2]$ implies $H_1 \le 6$; that $H_1 \le 3$ is equivalent to $H_1 = 2$ (the twin prime conjecture); that the minimal diameter of an admissible 3-tuple is exactly 6; and, via an explicit model, that DHL$[2,2]$ cannot be derived from the combinatorial form of DHL$[3,2]$ alone. The parity barrier for square-of-divisor-sum sieves enters only as an explicitly stated hypothesis. We do not prove DHL$[3,2]$ or the twin prime conjecture. All results are formalized in Lean 4 with Mathlib, without `sorry` and using only Lean's standard axioms.

1. INTRODUCTION

Let $p_1 = 2 < p_2 = 3 < \dots$ denote the primes and $H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n)$. For an admissible k-tuple $\mathcal{H}$ (Definition 2.1), the statement DHL$[k, j]$ asserts that infinitely many translates $n + \mathcal{H}$ contain at least j primes (Definition 2.2).

Following Goldston, Pintz and Yıldırım [2], Maynard [3] and the Polymath8b project [6], the unconditional bound $H_1 \le 246$ is known, and DHL$[3, 2]$ is known conditionally on the generalized Elliott–Halberstam conjecture (GEH), yielding $H_1 \le 6$ under GEH [6]. A claimed improvement to $H_1 \le 240$, attributed to a 2026 preprint of Stadlmann, is **[citation needed: unverified]** and is not used here.

The purpose of this note is to separate two statements that are sometimes conflated: DHL$[3, 2] \Rightarrow H_1 \le 6$ (Theorem 3.2), which is elementary, and $H_1 = 2$, the twin prime conjecture. Since every prime gap after $3 - 2$ is even (Theorem 3.1), $H_1 \le 3$ is equivalent to $H_1 = 2$, so reading DHL$[3, 2]$ as giving $H_1 \le 3$ is a category error.

By the widely accepted parity barrier (going back to Selberg), standard square-of-divisor-sum sieves are not expected to go below 6, even under GEH; in this note that barrier is used only as an explicit hypothesis (Hypothesis 3.5), not proved.

2. PRELIMINARIES

Definition 2.1. A finite set $\mathcal{H} \subset \mathbb{N}$ is *admissible* if for every prime q there is a residue class modulo q containing no element of $\mathcal{H}$.

Definition 2.2. For a predicate P on $\mathbb{N}$ and $k \ge j \ge 1$, let DHL$_P[k, j]$ be the statement: for every admissible $\mathcal{H}$ with $|\mathcal{H}| = k$ there are infinitely many n such that $\#\{h \in \mathcal{H} : P(n+h)\} \ge j$. Write DHL$[k, j]$ for DHL$_P[k, j]$ with $P =$ primality.

Definition 2.3. For $c \in \mathbb{N}$ write $H_1 \le c$ for the statement that $p_{n+1} - p_n \le c$ for infinitely many n. For an integer-valued sequence this is equivalent to $\liminf_n(p_{n+1} - p_n) \le c$.

Lemma 2.4. *The sets* $\{0, 2\}$, $\{0, 2, 6\}$ *and* $\{0, 4, 6\}$ *are admissible.*

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Proof. For $q \geq 7$ the elements are distinct residues below q and the class 1 is missed. For $q = 2$ the class 1 is missed. For $q = 3$: $\{0, 4, 6\} \equiv \{0, 1, 0\}$ misses 2; $\{0, 2, 6\} \equiv \{0, 2, 0\}$ and $\{0, 2\}$ miss 1. For $q = 5$: $\{0, 4, 6\} \equiv \{0, 4, 1\}$ misses 2; $\{0, 2, 6\} \equiv \{0, 2, 1\}$ misses 3; $\{0, 2\}$ misses 1. □

3. Main results

Theorem 3.1. *For all $n \geq 2$, $p_{n+1} - p_n \geq 2$ and $p_{n+1} - p_n$ is even. Consequently $H_1 \geq 2$, and $H_1 \leq 3$ if and only if $p_{n+1} - p_n = 2$ for infinitely many n.*

Proof. For $n \geq 2$, p_n and p_{n+1} are odd and distinct, so their difference is a positive even integer. Hence a gap ≤ 3 with $n \geq 2$ equals 2; the only gap equal to 1 is $p_2 - p_1$. □

Theorem 3.2. DHL$[3, 2]$ *implies* $H_1 \leq 6$.

Proof. By Lemma 2.4, $\{0, 4, 6\}$ is admissible, so DHL$[3, 2]$ gives infinitely many n such that at least two of $n, n+4, n+6$ are prime. For each such n we obtain primes $p < q$ with $q - p \in \{2, 4, 6\}$ and $p \geq n$. Write $p = p_k$. Since q is a prime larger than p, the next prime satisfies $p_{k+1} \leq q$, so $p_{k+1} - p_k \leq q - p \leq 6$. As $p \geq n$ and n is unbounded, so is k. □

Theorem 3.3. *Let $P(n)$ hold if and only if $n \bmod 4 \in \{0, 1\}$. Then* DHL$_P[3, 2]$ *holds (indeed for every 3-element set), but $P(n) \wedge P(n + 2)$ holds for no n; in particular* DHL$_P[2, 2]$ *fails.*

Proof. Integers differing by 2 have residues differing by 2 modulo 4, so they are never both in P. If $d \not\equiv 2 \pmod 4$, then $P(m) \wedge P(m + d)$ for all $m \equiv 0 \pmod 4$ when $d \equiv 0, 1$, and for all $m \equiv 1 \pmod 4$ when $d \equiv 3$. For $a < b < c$, the differences $b - a$ and $c - b$ cannot both be $\equiv 2 \pmod 4$, since then $c - a \equiv 0$. So some pair in the set has a difference $\not\equiv 2 \pmod 4$ and is hit infinitely often. Finally, $\{0, 2\}$ is admissible by Lemma 2.4. □

Remark 3.4. Theorem 3.3 shows only that no proof of DHL$[3, 2] \Rightarrow$ DHL$[2, 2]$ can rely on the combinatorial form of DHL$[3, 2]$ alone. It is *not* an independence result for the primes, and DHL$[3, 2]$ is compatible with $H_1 = 2$.

Hypothesis 3.5 (Parity barrier)**.** A given sieve method establishes DHL$[k, j]$ only if $j \leq \lceil k/2 \rceil$.

This hypothesis reflects the parity discussion in [6] for square-of-divisor-sum weights. It is assumed, not proved.

Theorem 3.6.

(1) *Under Hypothesis 3.5, a sieve method establishing* DHL$[k, 2]$ *has $k \geq 3$.*
(2) *(Unconditional.) No admissible 3-element set is contained in an interval $[a, a + 5]$. Together with Lemma 2.4, the minimal diameter of an admissible 3-set is exactly 6.*

Proof. (1) $2 \leq \lceil k/2 \rceil$ forces $k \geq 3$. (2) Admissibility modulo 2 forces the three elements to share a parity. If they lie in an interval of length 5, any two differ by 2 or 4, so they are pairwise distinct modulo 3 and cover all residues modulo 3, contradicting admissibility modulo 3. □

4. Formal verification

All results above are formalized in Lean 4 with Mathlib [5] in the repository `https://github.com/BenFrohman/DHL-3-2-implies-liminf-prime-gap-le-6`. The development contains no `sorry`; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. Primes are indexed from 0 in Lean ($p_0 = 2$). Key definitions and statements (rendered here in ASCII notation; the source uses Lean's Unicode symbols):

```
def H1Le (c : Nat) : Prop := forall N, exists n, n >= N /\ gap n <= c
def Admissible (H : Finset Nat) : Prop :=
  forall q, q.Prime -> exists r < q, forall h in H, h % q != r
def DHL (k j : Nat) : Prop :=
  forall H : Finset Nat, H.card = k -> Admissible H ->
    forall N, exists n, n >= N /\ j <= (H.filter (fun h => (n + h).Prime)).card

theorem dhl32_H1Le_six (h : DHL 3 2) : H1Le 6
theorem H1Le_three_iff_twin : H1Le 3 <-> forall N, exists n, n >= N /\ gap n = 2
theorem dhl32_not_implies_dhl22_combinatorially :
  exists (P : Nat -> Prop) (_ : DecidablePred P),
    DHLfor P 3 2 /\ (forall n, Not (P n /\ P (n + 2))) /\ Not DHLfor P 2 2
structure ParityBarrier (Proves : Nat -> Nat -> Prop) : Prop where
  bound : forall k j, Proves k j -> j <= (k + 1) / 2
```

Correspondence with the text:

- Lemma 2.4: `admissible_02`, `admissible_026`, `admissible_046`.
- Theorem 3.1: `two_le_gap`, `not_H1Le_one`, `H1Le_three_iff_two`, `H1Le_three_iff_twin`.
- Theorem 3.2: `dhl32_H1Le_six`.
- Theorem 3.3: `Ptoy_DHL32`, `dhl32_not_implies_dhl22_combinatorially`.
- Theorem 3.6: `ParityBarrier`, `parity_two_needs_three`, `admissible_three_diam_ge_six`, `min_diam_admissible_three`.

Statements from the literature (the bound 246, the conditional status of DHL[3, 2], and the unverified Stadlmann claim) are not formalized.

5. Discussion

A proof of $H_1 = 2$ must go beyond the combinatorial content of DHL[3, 2] (Remark 3.4) and, if it is sieve-theoretic, must overcome the limitation recorded in Hypothesis 3.5. Arguments that overcome or sidestep parity-type limitations exist in other settings. Friedlander and Iwaniec [1] showed that $X^2 + Y^4$ represents infinitely many primes, using bilinear (Type II) estimates to overcome the parity obstruction. Maynard [4] proved that there are infinitely many primes with a restricted digit, using Fourier-analytic control of Type I and Type II sums; this is a related use of bilinear-sum methods rather than a direct parity-breaking result of the Friedlander–Iwaniec kind. Until such an ingredient is brought to bear on prime gaps, DHL[3, 2] and $H_1 \leq 6$ remain the natural limit of the GPY/Maynard–Tao method under Hypothesis 3.5.

References

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